

P1040R10

std::embed

Published Proposal, 2026-05-12

This version:

https://thephd.dev/_vendor/future_cxx/papers/d1040.html

Author:

[JeanHeyd Meneide \(https://thephd.dev\)](https://thephd.dev)

Audience:

EWG, LEWG, CWG

Project:

ISO/IEC 14882 Programming Languages — C++, ISO/IEC JTC1/SC22/WG21

Latest:

https://thephd.dev/_vendor/future_cxx/papers/d1040.html

Implementation:

<https://github.com/ThePhD/embed>

Reply To:

[JeanHeyd Meneide](#)

Abstract

A proposal for a function that allows pulling resources at compile-time into a program, and a preprocessor directive to make finding those resources feasible for compilers and build systems.

Table of Contents

- 1 Revision History**
 - 1.1 Revision 10 - May 12th, 2026
 - 1.2 Revision 9 - March 25th, 2026
 - 1.3 Revision 8 - June 23rd, 2025
 - 1.4 Revision 7 - December 12th, 2024
 - 1.5 Revision 6 - March 2nd, 2020
 - 1.6 Revision 5 - January 13th, 2020
 - 1.7 Revision 4 - November 26th, 2018
 - 1.8 Revision 3 - November 26th, 2018
 - 1.9 Revision 2 - October 10th, 2018

1.10 Revision 1 - June 10th, 2018

1.11 Revision 0 - May 11th, 2018

2 Relevant Polls

2.1 DESIRED Polls

2.2 2026 Croydon: Evolution Working Group

2.3 Sofia 2025: Evolution Working Group

2.4 2020-2022

2.5 2018 Polls

3 Motivation

4 Scope and Impact

5 Implementation Experience & Current Practice

5.1 Speed Results

5.2 Memory Size Results

5.3 Results Analysis

6 Design Decisions

6.1 Primary Benefit: Recursive Processing and Composable String File Names

6.2 Prior Art

6.2.1 Literal-Based, `constexpr`

6.2.2 Literal-Based, Null Terminated (?)

6.2.3 Encoding

6.3 Design Goals

6.3.1 Implementation Defined

6.3.2 Binary Only

6.3.3 `constexpr` Compatibility

6.3.4 Modules

6.3.5 Statically Polymorphic

6.3.6 Optional Limit

6.3.7 Offset

6.3.8 Non-Unique / Shared Data

6.3.9 UTF-8 Only?

6.4 Dependency-Scanning Friendly with `#depend`

6.4.1 Implementation Quirk: Clang and `#depend` and the Preprocessor

6.4.2 Security: Globbs and Otherwise

7 Changes to the Standard

7.1 Intent

7.2 Proposed Feature Test Macro

7.3 Proposed Wording

7.3.1 Modify "§6.8.2 Object Model [**intro.object**]" one additional entry to the "... potentially non-unique object ..." list.

- 7.3.2 Modify "§15.1 Preamble [**cpp.pre**]" to add to the syntax
- 7.3.3 Append to "§15.12 Predefined macro names [**cpp.predefined**]" one additional entry
- 7.3.4 Add a new section "§15.5 Dependency [**cpp.depend**]"
- 7.3.5 Append to §17.1 General [**support.general**]'s **Table 38** one additional entry
- 7.3.6 Append to §17.3.1 General [**support.limits.general**]'s one additional entry
- 7.3.7 Add a new section "§17.✧ Constant Evaluation Resources [**support.res**]" in "§17 Language support [**support**]"

8 Appendix

- 8.1 Previous Design Discussion: Modules
- 8.2 Alternative
 - 8.2.1 Pre-Processing Tools Alternative
 - 8.2.2 `python` Alternative
 - 8.2.3 `ld` Alternative
 - 8.2.4 Manual Work
 - 8.2.5 The `incbin` tool
- 8.3 Previous Implementations
 - 8.3.1 `#depend` Soft Warnings, Hard Errors?
 - 8.3.2 String Table / Virtual File System

9 Acknowledgements

References

Informative References

§ 1. Revision History

§ 1.1. Revision 10 - May 12th, 2026

- Prepare for an EWG presentation at the request of the CWG chair by changing a significant portion (particularly the Tony Table and later paragraphs) of the [§ 3 Motivation](#).
- Once more point to existing implementation completed at ThePhD/embed GitHub repository and the existing patches for both GCC and Clang in [§ 5 Implementation Experience & Current Practice](#).
- Highlight that the recursive and similar bits of the proposal about their value have been talked about since 2019 in Belfast, shown in [§ 6.1 Primary Benefit: Recursive Processing and Composable String File Names](#).
- Mention that, as it always has been, it does not provide a unique address [§ 6.3.8 Non-Unique / Shared Data](#). Formalize this in the wording.
- Adjust wording to handle implementations which do partial preprocessing and do not fully honor the steps of a compiler such as Clang in [§ 6.4.1 Implementation Quirk: Clang and #depend and the Preprocessor](#).
- A presentation can (eventually, as the June 2026 meeting approaches) be found [here](#): https://thephd.dev/_presentations/standards/C++/2026%20June%20Brno/p1040/P1040%20-%20std%20embed.html.

§ 1.2. Revision 9 - March 25th, 2026

- Forward to CWG and LEWG from the Cryodon 2026 March meeting for C++29.
 - CWG to review the wording for core purposes.
 - LEWG to approve changes made to the function signature as mentioned in [§ 1.3 Revision 8 - June 23rd, 2025](#).
- Touch up some missing words in the [§ 7.1 Intent](#).
- Adjust and clean up wording, synopsis, and signatures in the [§ 7 Changes to the Standard](#) for LEWG and CWG review.
- CWG:
 - Completely overhaul the [§ 7.3 Proposed Wording](#).
 - Rewrite the specification for `#depend` to work appropriately with macro expansion and match the current specification for `#include/#embed`.
 - *header-name* is mapped to a sequence of Unicode code points, and then used to do matching to make it somewhat resistant to internal encoding shenanigans. (QUESTION: sequence of characters is better?)
 - Introduce a *input dependency check* algorithm. Currently, it relies on ECMA 262 to describe what is effectively globbing and recursive-globbing behavior. (Maybe it would be better to use a POSIX reference and talk about globbing instead?)
 - Stop italicizing defined terms.
 - Add wording for the non-uniqueness of the data returned from a `std::embed` call [§ 7.3.1 Modify "§6.8.2 Object Model \[intro.object\]" one additional entry to the "... potentially non-unique object ..." list](#).
 - Remove "Postconditions" from the library wording and replace with a better "Returns:" clause.
 - More appropriately define the backing storage of the `std::embed` call in the library wording.
- EWG changes to evaluate:
 - Whether or not we want data deduplication for the backing array of `std::embed` call returns (implied and assumed but never explicitly stated during discussion).
- LEWG changes to evaluate:
 - Whether or not `noexcept` should be removed to have `std::embed` throw an exception on file-not-found [§ 6.4.2 Security: Globs and Otherwise](#).
 - Whether or not we want `(u8/w)string_view` for `std::embed` or just back to the SG16 recommendation of [§ 6.3.9 UTF-8 Only?](#).
- Further discussion of modules has been moved to a new paper, [\[p1130\]](#).

§ 1.3. Revision 8 - June 23rd, 2025

- EWG settled on translation-unit specific dependencies; modules will be considered at a later date. See the vote [§ 2 Relevant Polls](#).
- Solidified wording for translation-unit based dependencies and removed relevant discussion in [§ 6.3.4 Modules](#).
- Some unfortunate implementation experience to accommodate the new `std::u8string_view/std::string_view/std::wstring_view` shenanigans in [§ 6.3.9 UTF-8 Only?](#).

- Added additional rationale for why `limit` and `offset` are present in §6.3.6 [Optional Limit](#) and §6.3.7 [Offset](#).
- Clarified that we are waiting for `bit_cast<...>(...)` to be improved as a means of better "compile time `reinterpret_cast`" rather than frontloading what `TYPE` in `std::embed<TYPE>(...)` is meant for in §6.3.5 [Statically Polymorphic](#).

§ 1.4. Revision 7 - December 12th, 2024

- Evaluate several different implementations and talk to several experts about the (now-closed) [\[p1130\]](#) with respect to `#depend`.
- Settle on `#depend` being exclusively based in translation units as this is both simple and easy to implement. Module export can be added on as a later paper, as described in §6.3.4 [Modules](#).
- Keep `#depend`'s glob syntax and reaffirm the speed of implementation is fine, and that "it's slow on Windows" is more or less poor programming practices on behalf of folks not willing to use modern Windows/NTFS APIs.
- Choose a specific option for how module behavior works (it is entirely private and local to the module).
- Provide `limit` and `offset` overloads, with `N` exact-limit version as well.

§ 1.5. Revision 6 - March 2nd, 2020

- Add new section §2 [Relevant Polls](#).
- Add new section §6.4 [Dependency-Scanning Friendly with #depend](#).
- Add new section §8.1 [Previous Design Discussion: Modules](#).
- Improve section §6.3.5 [Statically Polymorphic](#).
- Add new section §6.3.6 [Optional Limit](#).
- Add new section §6.3.9 [UTF-8 Only?](#).
- Add new section §8.3 [Previous Implementations](#).
- Improve wording and add static version (thanks, @lichray).

§ 1.6. Revision 5 - January 13th, 2020

- Split `#embed` into a new paper.
- Add memory and time benchmarks from various implementation strategies in the [new Current Practice section](#).
- Address concerns for a generic API and similar in the [new Results Analysis section](#).
- Retarget to EWG and SG 7.

§ 1.7. Revision 4 - November 26th, 2018

- Wording is now relative to [\[n4778\]](#).
- Minor typo and tweak fixes.

§ 1.8. Revision 3 - November 26th, 2018

- Change to using `constexpr`.
- Discuss potential issues with accessing resources after full semantic analysis is performed. Prepare to poll Evolution Working Group. Reference new paper, [p1130], about resource management.

§ 1.9. Revision 2 - October 10th, 2018

- Destroy `embed_options` and `alignment` options: if the function is materialized only at compile-time through `constexpr` or the upcoming "immediate functions" (`constexpr!`), there is no reason to make this part of the function. Instead, the user can choose their own alignment when they pin this down into a `std::array` or some form of C array / C++ storage.

§ 1.10. Revision 1 - June 10th, 2018

- Create future directions section, follow up on Library Evolution Working Group comments.
- Change `std::embed_options::null_terminated` to `std::embed_options::null_terminate`.
- Add more code demonstrating the old way and motivating examples.
- Incorporate LEWG feedback, particularly alignment requirements illuminated by Odin Holmes and Niall Douglass. Add a feature macro on top of having `__has_include( <embed> )`.

§ 1.11. Revision 0 - May 11th, 2018

- Initial release. 🚀

§ 2. Relevant Polls

The following polls are shaping the current design. Votes are in the form of SF (Strongly in Favor), F (in Favor), N (Neutral), A (Against), SA (Strongly Against).

§ 2.1. DESIRED Polls

These are the currently desired polls to move the paper forward.

P1040R10: LEWG prefers to use `size_t limit = numeric_limits<size_t>::max()` as a defaulted argument to `std::embed` to signal "read the whole file" as opposed to `optional<size_t> limit = nullopt`.

— SF F N A SA

— ∅ ∅ ∅ ∅ ∅

P1040R10: LEWG affirms the change to have 3 different `(u8/w)?string_view` overloads versus requiring exactly one overload of just UTF-8/`char8_t`.

— SF F N A SA

— 0 0 0 0 0

P1040R10: EWG would prefer that the storage for the static storage duration array backing the data returned by a `std::embed` call is non-unique (e.g., prefix-deduplicatable with different `std::embed` capable of having the same address just like string literals).

— SF F N A SA

— 0 0 0 0 0

§ 2.2. 2026 Croydon: Evolution Working Group

Forward P1040R8 (`std::embed` and `#depend`) to CWG and LEWG (to discuss the design of the interface+string types) for inclusion in C++29

— SF F N A SA

— 4 13 2 2 1

— Consensus. Reaffirmed support for translation unit dependencies only. Do module-based dependencies and potential `#depend export ...` in another paper (p1130).

§ 2.3. Sofia 2025: Evolution Working Group

P1040R7: EWG prefers using dependencies only from current translation unit meaning library wrapped `std::embed` in module won't work on user provided dependencies.

— SF F N A SA

— 4 4 6 1 0

— Consensus: translation unit dependencies only. Do module-based dependencies and potential `#depend export ...` in Primary Module Fragment in a different paper.

§ 2.4. 2020-2022

`#depend` - We would like to have this feature in C++(Something) and spend time figuring out the details.

— SF F N A SA

— 14 13 2 0 0

— Consensus: Do more work.

`#depend <foo/**>` - We want recursive globs (recursively searching through directories) for `#depend`.

— SF F N A SA

— 2 3 9 12 4

— Consensus: Do not want.

— Vote Commentary:

- A: Complexity?
- SA: Windows is slow with recursive globs.

It should be mandatory that EVERY file for `std::embed` is specified by a `#depend`.

- SF F N A SA
- 4 12 5 6 3
- Consensus: Split, no consensus. Add why/why not.
- Vote Commentary:
 - SF: This MUST exist. Both compiler and build system authors. (Implementers.)
 - SA: Can make user experience sad face for common case. Build system should scream at you for making the mistake instead.

`#depend` should form a Virtual File System / String Table State that constrains the search and should be passed to `std::embed`.

- SF F N A SA
- 3 11 7 2 2
- Consensus: Do it.
- Vote Commentary:
 - SA: Hell to implement. (This was the author.)

§ 2.5. 2018 Polls

Make `std::embed` ill-formed inside of a module interface (with a plan to revisit later).

- SF F N A SA
- 4 2 7 1 1
- Consensus: Yes, but Meh.
- Vote Commentary:
 - SA: Modules are important we should make sure it interacts well with modules (figure it out now).
 - SF: How does this work with `#depend` ?
 - SF: `std::embed` is basically a `#include` -- why would we want it in interface? Just focus on getting feature working and doing it well.
 - A: Jumping the gun. Space needs more exploration.
 - N: We are highly undecided - need to answer more questions (especially about Modules).

§ 3. Motivation

I'm very keen on `std::embed`. I've been hand-embedding data in executables for NEARLY FORTY YEARS now. — *Guy "Hatcat" Davidson, June 15, 2018*

The primary motivation is to allow for not only accessing data at compile-time, but providing seamless, optimizable data loading that can access that information and respond to that information **while embedding data based on what is read at constant evaluation time**. In particular, given two Lua files:

`core/base.lua`:

```
base = {
    ["value"] = 30
}
```

`main.lua`:

```
require 'core/base.lua'

function main()
    return 1 + base.value
end
```

A C++ program would be capable of doing this:

```
#include <span>
#include <phd/make_array.hpp>
#include <phd/embed.hpp>

int main(int, char*[]) {
    constexpr auto original_file_span = phd::embed<char>("resources/main.lua");

    std::printf("original file (%s):\n\n%s", "resources/main.lua", original_file.data());
    std::printf("\n\n=====

    constexpr auto original_file = comptime_make_string_array<original_file_span.size>(<orig
    constexpr auto final_glob = recursive_parse_lua("resources/main.lua");
    std::printf("`consteval recursive_parse_lua` result:\n\n%s", glob.data());

    return 0;
}
```

Without needing any additional build tool-based concatenation or partial-embedded with `#embed` before having those data globs engineered together through string table manipulation or -- worse -- runtime-based coordination such as with GLSL's `#include`-proxy system. Particularly, the output of the program would be:

```
original file ("resources/main.lua"):

require 'core/base.lua'
```

```
function main()
    return 1 + base.value
end
```

```
=====
`consteval recursive_parse_lua` result:
```

```
-- core/base.lua
base = {
    ["value"] = 30
}
```

```
function main()
    return 1 + base.value;
end
```

A very large amount of C and C++ programmer -- at some point -- attempts to `#include` large chunks of non-C++ data into their code. Of course, `#include` expects the format of the data to be source code, and thusly the program fails with spectacular lexer errors. Many different tools and practices were adapted to handle this, as far back as 1995 with the `xxd` tool. Many industries need such functionality, including (but hardly limited to):

- Financial Development
 - representing coefficients and numeric constants for performance-critical algorithms;
- Game Development
 - assets that do not change at runtime, such as icons, fixed textures and other data;
 - Shader and scripting code;
- Embedded Development
 - storing large chunks of binary, such as firmware, in a well-compressed format;
 - placing data in memory on chips and systems that do not have an operating system or file system;
- Application Development
 - compressed binary blobs representing data
 - non-C++ script code that is not changed at runtime;
- Server Development
 - configuration parameters which are known at build-time and are baked in to set limits and give compile-time information to tweak performance under certain loads;
 - SSL/TLS Certificates hard-coded into your executable (requiring a rebuild and potential authorization before deploying new certificates), and;
- Static Analyzers
 - Static analyzers suffer -- much like their binary code generating friends -- from having to parse extremely large array literals;

- Reduces memory pressure and enables better information tracking and potential sanitization (file source is not lost in build system).

In the pursuit of this goal, these tools have proven to have inadequacies and contribute poorly to the C++ development cycle as it continues to scale up for larger and better low-end devices and high-performance machines, bogging developers down with menial build tasks and trying to cover-up disappointing differences between platforms. It also absolutely destroys state-of-the-art compilers due to the extremely high memory overhead of producing an Abstract Syntax Tree for a braced initializer list of several tens of thousands of integral constants with numeric values at 255 or less.

The request for some form of `#include_string` or similar dates back quite a long time, with one of the oldest stack overflow questions asked-and-answered about it dating back nearly 10 years. Predating even that is a plethora of mailing list posts and forum posts asking how to get script code and other things that are not likely to change into the binary.

A simpler, C-friendly, preprocessor-only version of this functionality was standardized in C++26 and C23 with `#embed <file> params...`, which solved the problem for C and C-adjacent use cases. However, many of the use cases above -- Application Development, Server Development, Embedded Development, and Game Development in particular -- rely on files that rely on other files and feature some form of file-awareness for use of packaging and re-packing reusable bits of code. Whether it is `require '...'` in Lua or `#include <>` in HLSL, many of these data files rely on other data files in order to be complete: in this way, `#embed` is only useful for the simplest and easiest data files which are wholly uncomplicated and feature-complete on their own. Many, many more data formats and their resulting files need to be parsed and stitched together in part or in whole to be usable without a pre-determined build system step.

Additionally, the ability to validate data files at compile-time or -- even more powerfully -- rely on C++26 reflection to generate pitch-perfect interoperation functionality for Python, Go, Java, Lua, and more can all be achieved fairly powerfully if the C++ code could understand such files. The possibility to fully and completely cover much of the value proposition of SWIG, sol2, cons_expr, C-Python, JNI, and many other interface-generation driven by manual labor is fully tied to the ability to be able to on a basic level parse and handle interface declarations at compile-time and hang that off to further `constexpr`, `constexpr`, and `template` processing in C++.

This paper proposes `<embed>` to make this process much more efficient, portable, recursive, non-dependent on fixed preprocessor strings and streamlined.

§ 4. Scope and Impact

`std::embed` is an extension to the language proposed entirely as a library construct. The goal is to have it implemented with compiler intrinsics, builtins, or other suitable mechanisms. It does not affect the language. The proposed header to expose this functionality is `<embed>`, making the feature entirely-opt-in by checking if either the proposed feature test macro exists.

§ 5. Implementation Experience & Current Practice

There are a few cross-platform (and not-so-cross-platform) paths for getting data into an executable. We also scrutinized the performance, with numbers for both [memory overhead](#) and [speed overhead](#) available at the repository that houses the [current implementation](#). For ease of access, the numbers as of January 2020 with the latest versions of the indicated compilers and tools are replicated below.

Finally, a proper implementation has been completed in LLVM/Clang trunk and in GCC trunk as of April 25th, 2026, though they are not integrated into main/trunk of either compiler yet. They can be viewed [here](#):

<https://github.com/ThePhD/embed/tree/main/patches>. It can be played with online on [Godbolt](#).

§ 5.1. Speed Results

Below are timing results for a file of random bytes using a specific strategy. The file is of the size specified at the top of the column. Files are kept the same between strategies and tests.

- Intel Core i7-6700HQ @ 2.60 GHz
- 24.0 GB RAM 2952 MHz
- Debian Sid or Windows 10
- Method: Gather timings from [time](#) *nix program or [Measure-Command](#) { ... } PowerShell, compute mean

Strategy	4 bytes	40 bytes	400 bytes	4 kilobytes
#embed GCC	0.201 s	0.208 s	0.207 s	0.218 s
phd::embed GCC	0.709 s	0.724 s	0.711 s	0.715 s
xxd -generated GCC	0.225 s	0.215 s	0.237 s	0.247 s
xxd -generated Clang	0.272 s	0.275 s	0.272 s	0.272 s
xxd -generated MSVC	0.204 s	0.229 s	0.209 s	0.232 s
Circle @array	0.353 s	0.359 s	0.361 s	0.361 s
Circle @embed	0.199 s	0.208 s	0.204 s	0.368 s
objcopy (linker)	0.501 s	0.482 s	0.519 s	0.527 s

Strategy	40 kilobytes	400 kilobytes	4 megabytes	40 megabytes
#embed GCC	0.236 s	0.231 s	0.300 s	1.069 s
phd::embed GCC	0.705 s	0.713 s	0.772 s	1.135 s
xxd -generated GCC	0.406 s	2.135 s	23.567 s	225.290 s
xxd -generated Clang	0.366 s	1.063 s	8.309 s	83.250 s
xxd -generated MSVC	0.552 s	3.806 s	52.397 s	Out of Memory
Circle @array	0.353 s	0.363 s	0.421 s	0.585 s
Circle @embed	0.238 s	0.199 s	0.219 s	0.368 s
objcopy (linker)	0.500 s	0.497 s	0.555 s	2.183 s

Strategy	400 megabytes	1 gigabyte
#embed GCC	9.803 s	26.383 s
phd::embed GCC	4.170 s	11.887 s
xxd -generated GCC	Out of Memory	Out of Memory
xxd -generated Clang	Out of Memory	Out of Memory
xxd -generated MSVC	Out of Memory	Out of Memory
Circle @array	2.655 s	6.023 s
Circle @embed	1.886 s	4.762 s
objcopy (linker)	22.654 s	58.204 s

§ 5.2. Memory Size Results

Below is the peak memory usage (heap usage) for a file of random bytes using a specific strategy. The file is of the size specified at the top of the column. Files are kept the same between strategies and tests.

- Intel Core i7-6700HQ @ 2.60 GHz
- 24.0 GB RAM 2952 MHz
- Debian Sid or Windows 10
- Method: `/usr/bin/time -v` or Execute command hundreds of times, stare at Task Manager

Strategy	4 bytes	40 bytes	400 bytes	4 kilobytes
<code>#embed</code> GCC	17.26 MB	17.26 MB	17.26 MB	17.27 MB
<code>phd::embed</code> GCC	38.82 MB	38.77 MB	38.80 MB	38.80 MB
<code>xxd</code> -generated GCC	17.26 MB	17.26 MB	17.26 MB	17.27 MB
<code>xxd</code> -generated Clang	35.12 MB	35.22 MB	35.31 MB	35.88 MB
<code>xxd</code> -generated MSVC	< 30.00 MB	< 30.00 MB	< 33.00 MB	< 38.00 MB
Circle <code>@array</code>	53.56 MB	53.60 MB	53.53 MB	53.88 MB
Circle <code>@embed</code>	33.35 MB	33.34 MB	33.34 MB	33.35 MB
<code>objcopy</code> (linker)	17.32 MB	17.31 MB	17.31 MB	17.31 MB

Strategy	40 kilobytes	400 kilobytes	4 megabytes	40 megabytes
<code>#embed</code> GCC	17.26 MB	17.96 MB	53.42 MB	341.72 MB
<code>phd::embed</code> GCC	38.80 MB	40.10 MB	59.06 MB	208.52 MB
<code>xxd</code> -generated GCC	24.85 MB	134.34 MB	1,347.00 MB	12,622.00 MB
<code>xxd</code> -generated Clang	41.83 MB	103.76 MB	718.00 MB	7,116.00 MB
<code>xxd</code> -generated MSVC	~48.60 MB	~477.30 MB	~5,280.00 MB	Out of Memory
Circle <code>@array</code>	53.69 MB	54.73 MB	65.88 MB	176.44 MB
Circle <code>@embed</code>	33.34 MB	33.34 MB	39.41 MB	113.12 MB
<code>objcopy</code> (linker)	17.31 MB	17.31 MB	17.31 MB	57.13 MB

Strategy	400 megabytes	1 gigabyte
<code>#embed</code> GCC	3,995.34 MB	9,795.31 MB
<code>phd::embed</code> GCC	1,494.66 MB	5,279.37 MB
<code>xxd</code> -generated GCC	Out of Memory	Out of Memory
<code>xxd</code> -generated Clang	Out of Memory	Out of Memory
<code>xxd</code> -generated MSVC	Out of Memory	Out of Memory
Circle <code>@array</code>	1,282.34 MB	3,199.28 MB
Circle <code>@embed</code>	850.40 MB	2,128.36 MB
<code>objcopy</code> (linker)	425.77 MB	1,064.74 MB

§ 5.3. Results Analysis

The above clearly demonstrates the superiority of `std::embed` over latest optimized trunk builds of various compilers. It is also notable that originally the Circle language did not have an `@embed` keyword, but it was added in December 2019. When the compiler author was spoken to about Study Group 7's aspirations for a more generic way of representing data from a file, the ultimate response was this:

I'll add a new `@embed` keyword that takes a type and a file path and loads the file and embeds it into an array prvalue of that type. This will cut out the interpreter and it'll run at max speed. Feed back like this is good. This is super low-hanging fruit.

– [Sean Baxter, December 12th, 2019](#)

It was Circle's conclusion that a generic API was unsuitable and suffered from the same performance pitfalls that currently plagued current-generation compilers today. And it was SG7's insistence that a more generic API would be suitable, modeled on Circle's principles. Given that thorough exploration of the design space in Circle led to the same conclusion this proposal is making, and given the wide variety of languages providing a similar interface (D, Nim, Rust, etc.), it is clear that a more generic API is not desirable for functionality as fundamental and simple as this. This does not preclude a more generic solution being created, but it does prioritize the "Bird in the Hand" approach that the Direction Group and Bjarne Stroustrup have advocated for many times.

Furthermore, inspecting compiler bug reports around this subject area reveal that this is not the first time GCC has suffered monumental memory blowup over unoptimized representation of data. In fact, this is a [16+ year old problem that GCC has been struggling with for a long time now](#) (C++ version [here](#)). That the above numbers is nearing the best that can be afforded by some of the most passionate volunteers and experts curating an extremely large codebase should be testament to how hard the language is this area for compiler developers, and how painful it is for regular developers using their tools.

Clang, while having a better data representation and more optimized structures at its disposal, is similarly constrained. With significant implementation work, they are deeply constrained in what they can do:

It might be possible to introduce some sort of optimized representation specifically for initializer lists. But it would be a big departure from existing AST handling. And it wouldn't really open up new use cases, given that string literal handling is already reasonably efficient.

– [Eli Friedman, December 29th 2019](#)

Is this really the best use of compiler developer energy?

To provide a backdrop against which a big departure from current AST handling in can be compared, an implementation of the built-in necessary for this proposal is -- for an experienced developer -- at most a few day's work in either GCC or Clang. Other compiler engineers have reported similar ease of implementation and integration. Should this really be delegated to Quality of Implementation that will be need to be solved N times over by every implementation in their own particularly special way? Chipping away at what is essentially a fundamental inefficiency required by C++'s inescapable tokenization model from the preprocessor plus the sheer cost of an ever-growing language that makes simple constructs like a brace initializer list of integer constants expensive is, in this paper's demonstrated opinion, incredibly unwise.

§ 6. Design Decisions

`<embed>` avoids using the preprocessor or defining new string literal syntax like its predecessors, preferring the use of a free function in the `std` namespace. This gives `std::embed` a greater degree of power and advantage over `#embed`'s design is derived heavily from community feedback plus the rejection of the prior art up to this point, as well as the community needs demonstrated by existing practice and their pit falls.

§ 6.1. Primary Benefit: Recursive Processing and Composable String File Names

One of the biggest benefits of `std::embed` is that it's a normal, regular `constexpr` function call. This means that when processing data, it can be called recursively after reading the data from inside of the embedded file. This allows someone much greater degrees of freedom and power than is capable with either `#embed` or `_Pragma("embed ...")`, as it allows someone to use other constant expression parsing facilities along the introduction of file-based data. Among the primary use cases for this are:

- the ability to embed data recursively, such as processing sister language's that have their own `#include` or `import` statements like CG, HLSL, GLSL, etc.
- and, the ability to respond to the contents of the data and perform other compile-time data processing, such as search for certain records or performing deserialization or serialization.

Such behaviors can also be combined with the approach `meta` and Reflection facilities that are coming to C++, which would also allow for generative reflection in a set of files determined at constant evaluation time (rather than a fixed set at preprocessing time). This would enable to do things such as enable the generation of C++ code from data files or other languages, such as automatic binding layers to Lua tables/functions, Python objects/scripts/modules, and other robust integrations (such as Rust, JavaScript, etc.). Many of these languages have `require`, `include`, `import`, or other things which require multi-file coordination that `#embed` cannot handle.

This benefit has been discussed since the [November 2019 Belfast C++ Meeting](#) and at no point was it ever unambiguous what the fully-intentional reason for this proposal was:

`std::embed` Tradeoffs

- ✓ Works at constant evaluation time (right on time!)
- ✓ Implementable today (built-ins work just fine)
- ✓ Enables complex use cases not at all covered by non-programmable alternative
- ✗ Must copy out of span to separate storage to apply attributes.
- ✗ Not discoverable by ancient build tech (requires user to list (glob) dependency)

© Stephen's Oats, LLC & Embedded Memory 2019. All Rights Reserved.

§ 6.2. Prior Art

There has been a lot of discussion over the years in many arenas, from Stack Overflow to mailing lists to meetings with the Committee itself. The latest advancements that had been brought to WG21's attention was [p0373r0 - File String Literals](#). It

proposed the syntax `F"my_file.txt"` and `bF"my_file.txt"`, with a few other amenities, to load files at compilation time. The following is an analysis of the previous proposal. Additionally, some prior art and understanding comes from [#embed](#).

6.2.1. Literal-Based, constexpr

A user could reasonably assign (or want to assign) the resulting array to a `constexpr` variable as its expected to be handled like most other string literals. This allowed some degree of compile-time reflection. It is entirely helpful that such file contents be assigned to `constexpr`: e.g., string literals of JSON being loaded at compile time to be parsed by Ben Deane and Jason Turner in their CppCon 2017 talk, [constexpr All The Things](#).

6.2.2. Literal-Based, Null Terminated (?)

It is unclear whether the resulting array of characters or bytes was to be null terminated. The usage and expression imply that it will be, due to its string-like appearance. However, is adding an additional null terminator fitting for desired usage? From the existing tools and practice (e.g., `xxd -i` or linking a data-dumped object file), the answer is no: but the syntax `bF"hello.txt"` makes the answer seem like a "yes". This is confusing: either the user should be given an explicit choice or the feature should be entirely opt-in.

6.2.3. Encoding

Because the proposal used a string literal, several questions came up as to the actual encoding of the returned information. The author gave both `bF"my_file.txt"` and `F"my_file.txt"` to separate binary versus string-based arrays of returns. Not only did this conflate issues with expectations in the previous section, it also became a heavily contested discussion on both the mailing list group discussion of the original proposal and in the paper itself. This is likely one of the biggest pitfalls between separating "binary" data from "string" data: imbuing an object with string-like properties at translation time provide for all the same hairy questions around source/execution character set and the contents of a literal.

6.3. Design Goals

Because of the aforementioned reasons, it seems more prudent to take a "compiler intrinsic"/"magic function" approach. The function overload takes the form:

```
template <typename T = byte>
constexpr span<const T> embed(
    string_view resource_identifier,
    size_t offset = 0,
    optional<size_t> limit = std::nullopt
);

template <size_t N, typename T = byte>
constexpr span<const T, N> embed(
    string_view resource_identifier,
    size_t offset = 0
);
```



```

template <typename T = byte>
constexpr span<const T> embed(
    wstring_view resource_identifier,
    size_t offset = 0,
    optional<size_t> limit = std::nullopt
);

template <size_t N, typename T = byte>
constexpr span<const T, N> embed(
    wstring_view resource_identifier,
    size_t offset = 0
);

template <typename T = byte>
constexpr span<const T> embed(
    u8string_view resource_identifier,
    size_t offset = 0,
    optional<size_t> limit = std::nullopt
);

template <size_t N, typename T = byte>
constexpr span<const T, N> embed(
    u8string_view resource_identifier,
    size_t offset = 0
);

```

`resource_identifier` is a `string_view` processed in an implementation-defined manner to find and pull resources into C++ at constexpr time. `limit` is the maximum number of elements the function call can produce (but it may produce less). The most obvious source will be the file system, with the intention of having this evaluated as a core constant expression. We do not attempt to restrict the `string_view` to a specific subset: whatever the implementation accepts (typically expected to be a relative or absolute file path, but can be other identification scheme), the implementation should use. The `N` template parameter is for specifying that the returned span has *at least* `N` elements (to fit in the statically-sized `span`).

§ 6.3.1. Implementation Defined

Calls such as `std::embed( "my_file.txt" );`, `std::embed( "data.dll" );`, and `std::embed<vertex>( "vertices.bin" );` are meant to be evaluated in a `constexpr` context (with "core constant expressions" only), where the behavior is implementation-defined. The function has unspecified behavior when evaluated in a non-constexpr context (with the expectation that the implementation will provide a failing diagnostic in these cases). This is similar to how include paths work, albeit `#include` interacts with the programmer through the preprocessor.

There is precedent for specifying library features that are implemented only through compile-time compiler intrinsics (`type_traits`, `source_location`, and similar utilities). Core -- for other proposals such as [p0466r1 - Layout-compatibility and Pointer-interconvertibility Traits](#) -- indicated their preference in using a `constexpr` magic function implemented by intrinsic in the standard library over some form of `template <auto X> thing { /* implementation specified */ value; }` construct. However, it is important to note that [\[p0466r1\]](#) proposes type traits, where as this has entirely different functionality, and so its reception and opinions may be different.

Finally, we use "implementation defined" so that compilers can produce implementation-defined search path for their translation units and modules during compilation with flags. The current implementation uses `-fembed-`

`path=some/path/here/` to indicate this, but when it is standardized there will probably be an `-RI` or `-resource-include` flag instead. It is up to the implementation to pick what works for their platform. We rely on the existing machinery from the recently-standardized `#embed` and the recently changed wording, such that `std::embed` uses a "quote resource search".

§ 6.3.2. Binary Only

Creating two separate forms or options for loading data that is meant to be a "string" always fuels controversy and debate about what the resulting contents should be. The problem is sidestepped entirely by demanding that the resource loaded by `std::embed` represents the bytes exactly as they come from the resource. This prevents encoding confusion, conversion issues, and other pitfalls related to trying to match the user's idea of "string" data or non-binary formats. Data is received exactly as it is from the resource as defined by the implementation, whether it is a supposed text file or otherwise.

`std::embed( "my_text_file.txt" )` and `std::embed( "my_binary_file.bin" )` behave exactly the same concerning their treatment of the resource.

§ 6.3.3. Constexpr Compatibility

The entire implementation must be usable in a `constexpr` context. It is not just for the purposes of processing the data at compile time, but because it matches existing implementations that store strings and huge array literals into a variable via `#include`. These variables can be `constexpr`: to not have a `constexpr` implementation is to leave many of the programmers who utilize this behavior without a proper standardized tool.

§ 6.3.4. Modules

Per the EWG vote for direction in the § 2 Relevant Polls section, modules need more care and consideration and would need further consideration. There were two options presented to EWG:

1. (CHOSEN) Use the dependencies of the current translation unit.
2. (NOT CHOSEN) Accumulate dependencies from the root function invocation in the constant evaluation context, creating a stack of accessible `#depends` with each invoked function during constant evaluation.
3. (NOT CHOSEN) #2, but with the caveat that only `#depend export ...`s contribute to accessible depends and the others are "private" to the translation unit.

EWG chose to use the current translation unit. This is a simple, forwards-compatible choice while a separate paper will explore and nail down the semantics of use with modules. An early revision of [p1130r1] (revisions 0 and 1) tried to address this problem many, many years too early.

Now that modules have more well-defined semantics, the problem of dependency transfer can be more fully engaged with starting in [Revision 2](#) and further versions of [p1130].

§ 6.3.5. Statically Polymorphic

We allow for 3 different kinds of values in the returned `std::span` to be used: `std::byte`, `char`, and `unsigned char`. The last two are for compatibility with existing code. The first is for forward-facing designs that rely on `std::byte`. The template type provided defaults to `std::byte` to keep this forward-facing aspect of the design.

In the future, we expect that a `std::bit_cast` that uses arrays will move things forward for a compile-time way to take blobs of data and turn them into (potentially) all sorts of types, particularly ones which satisfy `std::is_trivial_v<TYPE>`.

6.3.6. Optional Limit

Consider some file-based resources that are otherwise un-sizeable and un-seek/tellable in various implementations such as `/dev/urandom`. Telling the compiler to go fetch data from this resource infinitely can result in compiler lockups or worse; therefore, the user can specify an additional parameter to the function call such as `std::embed("/dev/infinity", 0, 32);`. The 32 here is a `std::size_t limit` parameter in `std::embed`, and allows users to ask for up to but not more than `limit` elements in the returned `span`.

Note that as per §6.3.5 *Statically Polymorphic*, the limit is specified in terms of `T`s, not bytes. This means `sizeof(T) * CHAR_BIT` bits are required, and the implementation is mandated to require up to but not more than that many.

Additionally, a user can provide a template argument `N` of type `std::size_t` to return a `span<T, N>`. This requires at least `N` elements, but maybe more can exist. One can use both of these to request "exactly this many" elements, e.g. a call `std::embed<32>("/dev/infinity");` requires up to but not more than 32 elements (the `limit` parameter passed to the function) and that the returned span has at least 32 elements (the template parameter passed between the `<` and the `>`). Similarly, one could use `std::embed("/dev/infinity", 0, 32).first<32>();`. This fully covers the design space for what making a subset of the data is currently used for (e.g., prefixed data in a common model format that then has "back references" to the data contained in the first `n` elements).

6.3.7. Offset

There is also an `offset` parameter. While this can be achieved by calling `subspan(...)` on the returned value, it does not aid in how MUCH storage is stored on the code-behind and would require optimization to know that the edge(s) of the static storage data array to trim off. Having both `offset` and `limit` allow for the compiler itself to know how much data to retain *without* the use of optimization. It also allows a compiler frontend to merge potentially different calls which pull data offsets from various locations.

6.3.8. Non-Unique / Shared Data

The data that comes from `std::embed` is not unique. It is a matter of quality-of-implementation whether repeat calls with the exact same parameters produce the exact same data pointer. The data that comes from these types can point inside other static data. We make a small carve out in the wording, similar to that of String Literals, to allow for this.

6.3.9. UTF-8 Only?

This is related to a serious problem for string literals, particularly those of Translation Phase 7. When a user types a string literal such as `"Fáuncy.softłer"`, that string literal -- at Phase 5 of compilation -- gets translated from an idealized internal compiler character set to the "Execution Source Character Set". What this means is that the compiler is allowed to perform an implementation-defined translation from the string literal's ideal compiler representation to the execution character set (often, the presumed target execution character set). While this is not a problem for Clang -- which always uses UTF-8 -- and GCC -- which always uses UTF-8 unless someone specifically passes `-fexec-charset={something}` -- other com-

plers will take the string "Fáuncy.softler" and mangle it. This mangling does not have to come with warnings: in fact, MSVC will often times replace characters it cannot translate to the execution character set with either Mojibake or "?".

The solution that Study Group 16 recommended was to allow `std::u8string_view` and ONLY `std::u8string_view` as the parameter type. Others in SG-16 voiced concern that this would hamper general usability, but `u8` string literals and `char8_t` were put in the standard for reasons exactly such as this. The execution character set is an unknown and often lossy encoding on legacy systems: requiring UTF-8 with `std::u8string_view` matches most internal compiler representations of idealized text storage and provides a fairly representative way to work with the resource system.

Unfortunately, implementation experience attempting this found that there were a small subsection of files on all deployed operating systems (Windows, Linux-based, and MacOS (both normal and Linux through Asahi Linux)) that needed users to be precise about the byte sequence used to address storage on disk. As a backstop to the primary `std::u8string_view` template that we expect most users to employ with `u8""` string literals, we expect users to occasionally need to reach for `L""` (on fringe non-Windows and Windows machines) as well as `""` (to get an **exact** byte sequence to match specific platform peculiarities, like the absolute mess of file system issues that come from MacOS normalized vs. non-normalized file paths). We did not think this would be important in earlier revisions of the paper, but recently had to address the unfortunate reality of computing at the moment.

§ 6.4. Dependency-Scanning Friendly with `#depend`

One of the biggest hurdles to generating consensus was the deep-seated issues with dependency scanning. The model with only dealing with `#include` as a way of adding extra code or data-as-code means that all dependencies -- at the time of compiler invocation -- can be completely and statically known by the end of Phase 4 of compilation. This greatly aided in speedy dependency pulling. `std::embed` throws a wrench in that model as it can affect the generation of code at Phase 7 time, which is when `constexpr` is evaluated. By taking a `std::(u8)string_view`, it makes it impossible to know all files which may be used by a translation or module unit.

For this purpose, a new `#depend` preprocessor directive was introduced. It is intended to inform the compiler which translation units it depends on in order to make it simpler to retrieve all necessary dependencies. An advanced implementation can also replace all `#depend` directives with magical compiler builtins which instruct the compiler to cache the file's data as part of the translation unit. This makes it possible to send minimal compiler reproductions and test cases for bug vetting, as well as allow distributed systems to continue to use the `-fdirectives-only` or `-frewrite-includes` flags (or similar preprocessor flags).

The `#depend` directive works by allowing a user to specify a what are effectively globs that can be used later to filter out allowed calls to retrieve resources from `std::embed`:

```
// single-dependency directives
#depend <config/graph.bin>
#depend <foo.txt>
// family-dependencies
// do not "recurse" into directories
#depend "art/*"
#depend "art/mocks/*.json"
// recursive-family-dependency
// recurse through directories and similar
#depend "assets/**"
// mixed: all resources starting with
// "translation/", with all files that end in ".po",
```

```
// that have at least one "/" (one directory)
// after the "translation/", found recursively
#depend "translation/**/*.*.po"
```

Some blame was placed on Windows by stating that recursive directory iteration was too slow, and thus that `**` should not be part of the feature set (see § 2 Relevant Polls). However, the manifest existence of CMake's new `GLOB_RECURSE` in conjunction with `CONFIGURE_DEPENDS` very strongly nullifies the arguments brought up during the taking of that poll, which was that build systems could not do this fast enough on certain platforms (such as Windows). `GLOB_RECURSE` with `CONFIGURE_DEPENDS` was only allowed into CMake after benchmarks proved against conventional wisdom that this could be done swiftly. This was also backed up by the author of Boost's Low-Level File I/O library. Therefore, we feel compelled to ask the Committee to reconsider this now that there is sufficient data in the positive for such behavior.

`#depend` uses both the chevron-delimited `<foo.txt>` and the quote-delimited `"foo.txt"` formats. This is because one is for resources that are found only on the system / implementation paths similar to `-I` and `-isystem`, and then one that uses local look up plus the aforementioned resource paths. This can be useful for e.g. files that get embedded from system SDKs, like icons from Windows or Android build-time graphical resources or similar.

NOTE: It is also imperative to note that the way Include Directories and Embed Resource Directories work in all implementations is such that they work as-if by recursive glob. That is, `-I/mnt/code/boost-1.56.0/include` provides access to `<boost/soped_ptr.hpp>`, not just the singular folder `<boost>`. That is, every resource and every include underneath `boost` is allowed, such that they are effectively doing the equivalent of `#depend <boost/**>` to allow, recursively, for any file to be used. `#depend` is modeled off of that same access.

§ 6.4.1. Implementation Quirk: Clang and `#depend` and the Preprocessor

As specified by the standard, a full translation unit is preprocessed before the later stages of compilation are reached. Therefore, this program:

```
int main () {
    std::embed("foo.txt");
}

#depend "foo.txt"
```

is supposed to be fully legal. However, due to the quirks of how C and C++ work as well as the preprocessor, implementations do not actually have to preprocess input fully before moving on to parsing and type checking. Because there is currently no preprocessor directive which could possibly affect preceding preprocessing tokens or Phase 7+ elements, this program is supposed to be well-formed. Unfortunately, testing in Clang with large enough files triggered a failure if there was simply enough tokens before the `#depend` was encountered, resulting in spurious failures that were hard to track down where `std::embed("foo.txt")` would fail.

Therefore, the allowance of `#depend` that lexically follows its use from `std::embed` is conditionally supported. Separated preprocessors in the style of e.g. `[eElipsis]` or similar might work as expected, but implementations like Clang get to choose whether or not. The reason we do this is because it would be a serious change in Clang's implementation to let this work, even if technically -- by the description of the standard -- it should.

§ 6.4.2. Security: Globbs and Otherwise

It is important to note that, insofar as security is concerned, `#depend` that use glob-style syntax is **not** what triggers the full consumption of files from the file system, nor their transport to e.g. a CI server in order to make using `std::embed` on remote conglomerate build servers and similar work. `#depend` is a **whitelist** of files: it does not put any data into the program. You cannot "glob" files into the program, and there is no file system search capabilities built into this program: if a file is not found, the program is **ill-formed**, which is different from finding a file that is empty (in which case the returned `std::span<T>` is empty).

Additionally, security prospects are slightly improved by the fact that a failure to find a resource results in an ill-formed program. This means that a bad actor cannot continuously probe for a set of files until it finds one at build-time; any failed search stops compilation. This is not a fool-proof security or defense measure, but provides a slight check against code that would see to simply slurp up the contents of a file system. However, this does not stop an attacker from simply guessing all of the right files correctly, so it is only a small mitigation. This problem is better solved by build system sandboxing and proper data handling practices.

Either way, if this is not deemed as important enough, it is completely viable to change the ill-formed in the future to a thrown `constexpr` exception, now that C++26 allows for those. We would leave this determination to a future paper after it considers the arguments for and against such compile-time programmability.

Finally, the additional security comes from the use of an anchoring `$` at the end of the "dependency pattern" that is used to perform the "input dependency check". This does not allow arbitrary paths that just "happen" to include the depended-on pattern: each `#depend` is anchored as being responsible for describing the tail end of a file. This in conjunction with the `--embed-dir=` resource-finding flags of existing implementations is enough to keep the directory use well-scoped.

NOTE: If an attacker has control of your build flags to make this a problem, you have MUCH bigger problems than the behavior of `std::embed`.

§ 7. Changes to the Standard

Wording changes are relative to the latest working draft.

§ 7.1. Intent

The intent of the wording is to provide a function that:

- handles the provided resource-identifying `string_view/u8string_view/wstring_view` in an implementation-defined manner;
- retrieves the data as-if using `std::fread` on a file opened at constant expression time;
- fills the data in appropriately, up to a limit.
- and, returns the specified `constexpr span` representing either the bytes of the resource or the bytes view as the type `T`.

The wording also explicitly disallows the usage of the function outside of a core constant expression by marking it `constexpr`, meaning it is ill-formed if it is attempted to be used at not-constexpr time (`std::embed` calls should not show up as a function in the final executable or in generated code). The program may pin the data returned by `std::embed` through the span into the executable if it is used outside a core constant expression.

For `#depend`, the purpose is to provide an implementation-defined set of search paths that get used only for `std::embed` later. This makes the specification lighter than generally anticipated. Because `#depend` is private to any given module, we simply do not touch the existing specification and thusly the preprocessor directive remains private in its effects. We explicitly do not add it to the wording around macro imports or *pp-imports*.

§ 7.2. Proposed Feature Test Macro

The proposed feature test macros are `__cpp_lib_embed` for the library and `__cpp_pp_depend` for the preprocessor functionality.

§ 7.3. Proposed Wording

§ 7.3.1. Modify "§6.8.2 Object Model [intro.object]" one additional entry to the "... potentially non-unique object ..." list.

...

- 9 An object is a potentially non-unique object if it is
 - a string literal object (5.13.5 [[lex.string](#)]),
 - the backing array of an initializer list (9.5.4 [[dcl.init.ref](#)]), or
 - the backing array pointed to by the returned value from a `embed` call (17.✱.1 [[support.res.embed](#)]), or
 - a template parameter object of array type (21.4.3 [[meta.define.static](#)]), or
 - a subobject thereof.

...

7.3.2. Modify "\$15.1 Preamble [cpp.pre]" to add to the syntax

```
...  
  
control-line:  
  
    # include pp-tokens new-line  
  
    pp-import  
  
    # embed pp-tokens new-line  
  
    # depend pp-tokens new-line  
  
    # define identifier replacement-list new-line  
  
    # define identifier lparen identifier-listopt ) replacement-list new-line  
  
    # define identifier lparen ... ) replacement-list new-line  
  
    # define identifier lparen identifier-list , ... ) replacement-list new-line  
  
    # undef identifier new-line  
  
line-directive  
  
    # error pp-tokensopt new-line  
  
    # warning pp-tokensopt new-line  
  
    # pragma pp-tokensopt new-line  
  
    # new-line  
  
...
```

7.3.3. Append to "\$15.12 Predefined macro names [cpp.predefined]" one additional entry

```
#define __cpp_pp_depend ????? /*  NOTE: EDITOR VALUE HERE */
```


15.5 Dependency [cpp.depend]

- 1 A `#depend` directive provides a dependency pattern to be used for performing input dependency checks. A *dependency patterns* is a sequence of Unicode code points. *Input dependencies* are the collection of dependency patterns accumulated by `#depend` directives throughout a translation unit.

- 2 A preprocessing directive of the form

```
# depend header-name new-line
```

adds a dependency pattern to the input dependencies. After removal of the surrounding paired " (U+0022 QUOTATION MARK), or paired < (U+003C LESS-THAN SIGN) and > (U+003D GREATER-THAN SIGN), the mapping of the rest of a *header-name* to the sequence of Unicode code points representing the dependency pattern is implementation-defined.

- 3 A preprocessing directive of the form

```
# depend pp-tokens new-line
```

(that does not match the previous form) is permitted. The preprocessing tokens after `depend` in the directive are processed just as in normal text (i.e., each identifier currently defined as a macro name is replaced by its replacement list of preprocessing tokens). The resulting sequence of preprocessing tokens shall be of the form

```
header-name-tokens
```

An attempt is then made to form a header-name preprocessing token ([lex.header]) from the whitespace and the characters of the spellings of the header-name-tokens; the treatment of whitespace is implementation-defined. If the attempt succeeds, the directive with the so-formed header-name is processed as specified for the previous form. Otherwise, the program is ill-formed, no diagnostic required.

[Note: Adjacent *string-literals* are not concatenated into a single *string-literal* (see the translation phases in [lex.phases](#)); thus, an expansion that results in two *string-literals* is an invalid directive. — end note]

- 4 An *input dependency check* takes an input that is a sequence of Unicode code points and yields either success or failure according to the following matching algorithm. For each dependency pattern, its sequence of Unicode code points is converted to an Standard Ecma 262 (called ECMA-262 in this clause) regular expression pattern (see ECMA-262 subclause 15.10) in the following way.

For each Unicode code point in the dependency pattern:

- If it is a * (U+002A ASTERISK) and:
 - if it is followed by another * (U+002A ASTERISK), then the two asterisks are replaced by the ECMA-262 regular expression *Term* .* (U+002E FULL STOP, U+002A ASTERISK).
 - Otherwise, if it not followed by another * (U+002A ASTERISK), then the single asterisk is replaced by the ECMA-262 regular expression *CharacterClass* of [*^path-separators*]* (U+005B LEFT SQUARE BRACKET, U+005E CIRCUMFLEX ACCENT, *path-separators*, U+005D RIGHT SQUARE BRACKET, U+002A ASTERISK), where *path-separators* are the appropriate ECMA-262 regular expression *ClassAtoms* for a slash (U+002F SOLIDUS), a backslash (U+005C REVERSE SOLIDUS), or any other corresponding implementation-defined directory-separator ([fs.path.generic](#)).

- Otherwise, it is replaced by an appropriate ECMA-262 regular expression *Atom* representing the Unicode code point of the sequence.

Finally, the ECMA-262 regular expression *CharacterClass* \$ (U+0024 DOLLAR SIGN) is appended to the pattern.

[*Note*— The *ClassAtom* and *Atom* placed into these regular expressions representing a plain Unicode code point appropriately escapes any Unicode code point that would otherwise be interpreted as non-*ClassAtom* or non-*Atom* in the overall regular expression. — *end Note*]

The input is then matched against the constructed ECMA-262 regular expression using the algorithm specified in ECMA-262 15.10. If the input matches the converted dependency pattern, then the input dependency check yields success. Otherwise, the input is checked against the next dependency pattern as described previously. If all dependency patterns in the input dependencies are checked against the input and none yield success, then the input dependency check yields failure.

- 5 *Recommended practice*: Implementations should use the input dependencies to inform the potential cache of state for the translation environment when it is beneficial, such as for proper replication of an environment for translation of programs prior to execution in non-local environment.

7.3.5. Append to §17.1 General [support.general]'s Table 38 one additional entry

Subclause	Header(s)
...	...
17.✱	Resources <embed>

7.3.6. Append to §17.3.1 General [support.limits.general]'s one additional entry

```
#define __cpp_lib_embed ????? /* NOTE: EDITOR VALUE HERE */
```

§ 7.3.7. Add a new section "§17.✧ Constant Evaluation Resources [support.res]" in "§17 Language support [support]"

17.✧ Constant Evaluation Resources [support.res]

17.✧.1 Header <embed> synopsis [support.embed.syn]

```
#include <string_view> // see [string_view.syn]

namespace std {
    template<class T = byte>
    constexpr span<const T> embed(string_view resource_identifier,
        size_t offset = 0,
        size_t limit = numeric_limits<size_t>::max()) noexcept;

    template<size_t N, class T = byte>
    constexpr span<const T, N> embed(string_view resource_identifier,
        size_t offset = 0) noexcept;

    template<class T = byte>
    constexpr span<const T> embed(wstring_view resource_identifier,
        size_t offset = 0,
        size_t limit = numeric_limits<size_t>::max()) noexcept;

    template<size_t N, class T = byte>
    constexpr span<const T, N> embed(wstring_view resource_identifier,
        size_t offset = 0) noexcept;

    template<class T = byte>
    constexpr span<const T> embed(u8string_view resource_identifier,
        size_t offset = 0,
        size_t limit = numeric_limits<size_t>::max()) noexcept;

    template<size_t N, class T = byte>
    constexpr span<const T, N> embed(u8string_view resource_identifier,
        size_t offset = 0) noexcept;
}
```

17.✧.2 Function template embed [support.embed]

```
namespace std {
    template<class T = byte>
    constexpr span<const T> embed(string_view resource_identifier,
        size_t offset = 0,
        size_t limit = numeric_limits<size_t>::max()) noexcept;

    template<size_t N, class T = byte>
    constexpr span<const T, N> embed(string_view resource_identifier,
        size_t offset = 0) noexcept;

    template<class T = byte>
    constexpr span<const T> embed(wstring_view resource_identifier,
        size_t offset = 0,
```

```

    size_t limit = numeric_limits<size_t>::max()) noexcept;

template<size_t N, class T = byte>
constexpr span<const T, N> embed(wstring_view resource_identifier,
    size_t offset = 0) noexcept;

template<class T = byte>
constexpr span<const T> embed(u8string_view resource_identifier,
    size_t offset = 0,
    size_t limit = numeric_limits<size_t>::max()) noexcept;

template<size_t N, class T = byte>
constexpr span<const T, N> embed(u8string_view resource_identifier,
    size_t offset = 0) noexcept;
}

```

- 1 The sequence of characters represented by `resource_identifier` is used to perform a quote resource search for a resource (15.4 [cpp.embed]). The conversion of the sequence of characters represented by `resource_identifier` to an appropriate sequence of characters for a quote resource search is implementation-defined. If the implementation cannot find the resource identified by the `resource_identifier` after exhausting the sequence of implementation-defined search locations, or if the implementation finds the resource specified but that same resource's `resource_identifier` fails an input dependency check (15.5 [cpp.depend]), then the program is ill-formed outside of any immediate context. The mapping of the sequence of characters represented by `resource_identifier` to an appropriate sequence of Unicode code points for use with the input dependency check is implementation-defined.

- 2 Let *implementation-resource-width* be the implementation-defined width of the resource. Let *implementation-resource-count* be `implementation-resource-width / CHAR_BIT`. Let *resource-offset* be `min(offset, implementation-resource-count)`.

Let *resource-limit* be:

- `limit` if an overload with the `limit` parameter is used.
- `N` if an overload with the template parameter `N` is used.
- Otherwise, `implementation-resource-count`.

Let *resource-count* be `max(0, min(implementation-resource-count - resource-offset, resource-limit))`.

- 3 *Mandates:*

- `T` is one of `byte`, `char`, or `unsigned char`.
- If an overload with the template parameter `N` is used, then `resource-count` is at least `N`.
- `Implementation-resource-width` is an integral multiple of `CHAR_BIT`.

- 4 *Returns:* A `span<const T, N>(p, s)` for overloads with the template parameter `N`, otherwise a `span<const T>(p, s)`, where

- `s` is `resource-count`.

§ 8. Appendix

§ 8.1. Previous Design Discussion: Modules

Due to the vote taken in [§ 2 Relevant Polls](#) at Revision 7 of this paper, the question posed by this section shown in this section was answered as "translation units only". The hope is to move on with translation units right now per EWG direction and then revisit this question in a separate paper, which will give these details and concerns the answers they deserve.

Modules front-load and bring front-and-center one of the largest problems not with `std::embed`, but with `#depend`. `#depend` is a Compilation Phase 4 entity: that is, it does not exist beyond the preprocessor. In the world of `#include`, preprocessor commands were resolved that would not exist in the traditional `#include` world, or at least would not have to need immediate answers. In particular, the fact that modules can "save" Phase 4 information before proceeding with the rest of compilation opens interesting questions for Module Interface Units (AKA, the public-facing portion of modules). Module implementation units and other module-related bits are, thankfully, entirely unaffected.

The problem is illustrated most powerfully by the following snippet:

`m0.C++m:`

```
export module m0;

#depend "header.bin"

import <embed>;
import combine;

export consteval auto f0(std::string_view f) {
    const auto h = std::embed("header.bin");
    const auto t = std::embed(f);
    return combine(h, t);
}
```

`m1.C++m:`

```
export module m1;

#depend "default.stuff"

import m0;

export consteval auto f1(std::string_view f = "default.stuff") {
    return f0(f);
}
```

```
export consteval auto getpath() {
    return "default.stuff";
}
```

translation_unit0.c++:

```
import m1;
import print;
import <embed>

#depend "coolstuff.bin"

int main() {
    print(f1("coolstuff.bin")); // [0] fails
    print(f1());                // [1] fails
    std::embed("header.bin");    // [2] fails
    std::embed(getpath());       // [3] fails
    std::embed("coolstuff.bin"); // [4] ok
}
```

All of [0], [1], [2], and [3] fail because each of them is trying to access dependencies that are part of translation units outside of the current translation unit. That is: modules do not transfer their dependencies outside of themselves. This seems unnecessarily cruel and hostile, but behaves in this manner to keep the specification simple and implementable. In the future or perhaps in another revision of this paper, additional work can be done to define the relationship in terms of translation units under a module accumulating all of their `#depends`s together. There is also some speculation on making such dependencies explicit with an additional form of `#depend` like:

deps.c++m:

```
export module deps;

#depend <sdk/private/**>
#depend export <sdk/everything/**>
```

main.cpp:

```
import deps;

int main () {
    std::embed("sdk/everything/meow.wav"); // ok
    std::embed("sdk/private/super_secret_sauce.mix"); // ill-formed
    return 0;
}
```

This would make the behavior opt-in and predictable. We do not include it in the most recent revision of this paper and plan to talk to SG15 and Modules Experts to craft this wording in a different version of this paper, after getting approval for the core feature and feedback from EWG. We also need to resolve the behavior for `cpp.import`-style header-unit importing. As of right now, no wording is provided for this section with respect to `std::embed` and resource dependencies, so as of

right now importing a module versus importing a header unit does not change much insofar as our limited understanding is concerned.

§ 8.2. Alternative

Other techniques used include pre-processing data, link-time based tooling, and assembly-time runtime loading. They are detailed below, for a complete picture of today's sad landscape of options.

§ 8.2.1. Pre-Processing Tools Alternative

1. Run the tool over the data (`xxd -i xxd_data.bin > xxd_data.h`) to obtain the generated file (`xxd_data.h`):

```
unsigned char xxd_data_bin[] = {
    0x48, 0x65, 0x6c, 0x6c, 0x6f, 0x2c, 0x20, 0x57, 0x6f, 0x72, 0x6c, 0x64,
    0x0a
};
unsigned int xxd_data_bin_len = 13;
```

1. Compile `main.cpp`:

```
#include <iostream>
#include <string_view>

// prefix as constexpr,
// even if it generates some warnings in g++/clang++
constexpr
#include "xxd_data.h"
;

template <typename T, std::size_t N>
constexpr std::size_t array_size(const T (&)\[N]) {
    return N;
}

int main() {
    static_assert(xxd_data_bin[0] == 'H');
    static_assert(array_size(xxd_data_bin) == 13);

    std::string_view data_view(
        reinterpret_cast<const char*>(xxd_data_bin),
        array_size(xxd_data_bin));
    std::cout << data_view << std::endl; // Hello, World!
    return 0;
}
```

Others still use python or other small scripting languages as part of their build process, outputting data in the exact C++ format that they require.

There are problems with the `xxd -i` or similar tool-based approach. Lexing and Parsing data-as-source-code adds an enormous overhead to actually reading and making that data available.

Binary data as C(++) arrays provide the overhead of having to comma-delimit every single byte present, it also requires that the compiler verify every entry in that array is a valid literal or entry according to the C++ language.

This scales poorly with larger files, and build times suffer for any non-trivial binary file, especially when it scales into Megabytes in size (e.g., firmware and similar).

8.2.2. `python` Alternative

Other companies are forced to create their own ad-hoc tools to embed data and files into their C++ code. MongoDB uses a [custom python script](#), just to get their data into C++:

```
import os
import sys

def jsToHeader(target, source):
    outFile = target
    h = [
        '#include "mongo/base/string_data.h"',
        '#include "mongo/scripting/engine.h"',
        'namespace mongo {',
        'namespace JSFiles{',
    ]
    def lineToChars(s):
        return ','.join(str(ord(c)) for c in (s.rstrip() + '\n')) + ','
    for s in source:
        filename = str(s)
        objname = os.path.split(filename)[1].split('.')[0]
        stringname = '_jsgcode_raw_' + objname

        h.append('constexpr char ' + stringname + "[] = {")

        with open(filename, 'r') as f:
            for line in f:
                h.append(lineToChars(line))

        h.append("0};")
        # symbols aren't exported w/o this
        h.append('extern const JSFile %s;' % objname)
        h.append('const JSFile %s = { "%s", StringData(%s, sizeof(%s) - 1) };' %
            (objname, filename.replace('\\', '/'), stringname, stringname))

    h.append("} // namespace JSFiles")
    h.append("} // namespace mongo")
    h.append("")

    text = '\n'.join(h)

    with open(outFile, 'wb') as out:
```

```

try:
    out.write(text)
finally:
    out.close()

if __name__ == "__main__":
if len(sys.argv) < 3:
    print "Must specify [target] [source] "
    sys.exit(1)
jsToHeader(sys.argv[1], sys.argv[2:])

```

MongoDB were brave enough to share their code with me and make public the things they have to do: other companies have shared many similar concerns, but do not have the same bravery. We thank MongoDB for sharing.

8.2.3. ld Alternative

A full, compilable example (except on Visual C++):

0. Have a file `ld_data.bin` with the contents `Hello, World!`.
1. Run `ld -r binary -o ld_data.o ld_data.bin`.
2. Compile the following `main.cpp` with `c++ -std=c++17 ld_data.o main.cpp`:

```

#include <iostream>
#include <string_view>

#ifdef __APPLE__
#include <mach-o/getsect.h>

#define DECLARE_LD(NAME) extern const unsigned char _section$__DATA_##NAME[];
#define LD_NAME(NAME) _section$__DATA_##NAME
#define LD_SIZE(NAME) (getsectbyname("__DATA", "__" #NAME)->size)

#elif (defined __MINGW32__) /* mingw */

#define DECLARE_LD(NAME) \
    extern const unsigned char binary_##NAME##_start[]; \
    extern const unsigned char binary_##NAME##_end[];
#define LD_NAME(NAME) binary_##NAME##_start
#define LD_SIZE(NAME) ((binary_##NAME##_end) - (binary_##NAME##_start))

#else /* gnu/linux ld */

#define DECLARE_LD(NAME) \
    extern const unsigned char _binary_##NAME##_start[]; \
    extern const unsigned char _binary_##NAME##_end[];
#define LD_NAME(NAME) _binary_##NAME##_start
#define LD_SIZE(NAME) ((_binary_##NAME##_end) - (_binary_##NAME##_start))
#endif

```

```

DECLARE_LD(ld_data_bin);

int main() {
    // impossible
    //static_assert(xxd_data_bin[0] == 'H');
    std::string_view data_view(
        reinterpret_cast<const char*>(LD_NAME(ld_data_bin)),
        LD_SIZE(ld_data_bin)
    );
    std::cout << data_view << std::endl; // Hello, World!
    return 0;
}

```

This scales a little bit better in terms of raw compilation time but is shockingly OS, vendor and platform specific in ways that novice developers would not be able to handle fully. The macros are required to erase differences, lest subtle differences in name will destroy one's ability to use these macros effectively. We omitted the code for handling VC++ resource files because it is excessively verbose than what is present here.

N.B.: Because these declarations are `extern`, the values in the array cannot be accessed at compilation/translation-time.

§ 8.2.4. Manual Work

Many developers also hand-wrap their files in (raw) string literals, or similar to massage their data -- binary or not -- into a conforming representation that can be parsed at source code:

0. Have a file `data.json` with some data, for example:

```
{ "Hello": "World!" }
```

1. Mangle that file with raw string literals, and save it as `raw_include_data.h`:

```
R"json({ "Hello": "World!" })json"
```

1. Include it into a variable, optionally made `constexpr`, and use it in the program:

```

#include <iostream>
#include <string_view>

int main() {
    constexpr std::string_view json_view =
#include "raw_include_data.h"
    ;

    // { "Hello": "World!" }
    std::cout << json_view << std::endl;
    return 0;
}

```

This happens often in the case of people who have not yet taken the "add a build step" mantra to heart. The biggest problem is that the above C++-ready source file is no longer valid in as its original representation, meaning the file as-is cannot be passed to any validation tools, schema checkers, or otherwise. This hurts the portability and interop story of C++ with other tools and languages.

Furthermore, if the string literal is too big vendors such as VC++ will hard error [the build \(example from Nonius, benchmarking framework\)](#).

§ 8.2.5. The `incbin` tool

There is a tool called `[incbin]` which is a 3rd party attempt at pulling files in at "assembly time". Its approach is incredibly similar to `ld`, with the caveat that files must be shipped with their binary. It unfortunately falls prey to the same problems of cross-platform woes when dealing with VC++, requiring additional pre-processing to work out in full.

§ 8.3. Previous Implementations

This section is primarily to address feedback from polls wherein different forms and implementation strategies were asked for by the Evolution Working Group and other implementers. A tour of the design and implementation these cases helps show what has been considered.

§ 8.3.1. `#depend` Soft Warnings, Hard Errors?

The current specification makes it a hard error if a file has not been identified previously by a `#depend` directive. This makes the simple case of including a single file a bit more annoying than it needs to be, but also makes the case of general development with a non-distributed build system a pain. To be perfectly transparent, the author of this paper and almost 100% of the author's clients do not use distributed anything to build their code: errors at "file not found" are generally useful enough. Making the lack of matching `#depend` a hard error seems like pushing an important but nevertheless Committee-over-represented concern (i.e. distributed build tool users) onto all C++ users.

While the author feels a lot better about it being a soft warning that can be turned into a hard error by use of `-Werror-depend` for distributed build users, we leave the specification to strongly encourage implementations to hard error on an inability to not only find the resource but match it with a previous `#depend` declaration.

§ 8.3.2. String Table / Virtual File System

This implementation idea was floated twice, once during SG-7 discussion at the November 2019 Belfast meeting and again during the February 2020 Prague meeting. The crux of this version of `std::embed` is that it does not take resource identifiers related to e.g. the file system directly: a directive is used to build a "String Table" or "Translation Table" from system-specific paths to "friendly" names:

```
// map "foo.txt" to file name "foo"
#depend_name "foo.txt" "foo"

#ifdef _WIN32
    // map Windows-specific resource
    // "win/bazunga.bin" to file name "baz"
```

```

    #depend_name "win/bazunga.bin" "baz"
#else
    // map Unix-specific resource
    // "nix/bazooka.bin" to file name "baz"
    #depend_name "nix/bazooka.bin" "baz"
#endif

#include <embed>

int main () {
    // pulls foo.txt
    constexpr std::span<std::byte> foo = std::embed("foo");
    // pulls either bazunga or bazooka
    constexpr std::span<std::byte> baz = std::embed("baz");
    return foo[0] == 'f' && baz[2] == '\x3';
}

```

On the tin, this seems to bring nice properties to the table. We get to "erase" platform-specific paths and give them common names, we have a static list of names that we always pull from, and more. However, there are several approaches to this problem. Consider one of the primary use cases for `std::embed` as a `constexpr` function: reading a resource and, based on its contents, `std::embed`-ing other resources.

This becomes a problem: if `foo.txt` has a `get: bar.doc` line of text in it, we read that line from `foo.txt`, and then attempt to `std::embed("bar.doc")`, we get an error because we did not give `bar.doc` a string table name. This means we need to go back and not only mark it as a dependency with `#depend`, but also give it a static string-table based name.

Even conquering that problem (with, e.g., glob-based `#depend` families of dependencies), we face another: resource files -- JSON, SPIR-V, Wavefront OBJ, HLSL/GLSL, python, Lua, HSAIL, whatever -- do not speak "C++ String Table" to name their identifiers. Generally, these speak "File System": we are adding a level of indirection that makes it difficult to keep working with the file system, especially when it comes to working with external resources. Most tools communicate and traffic interchange information VIA URIs or relative / local file paths. Forcing users to either adapt their code to play nice with this new C++ file system or maintaining a translation table to and from "C++ String Table" to "C++ File System" is an order of magnitude additional complexity.

There is absolutely room for a (potentially `constexpr`) Virtual File System in C++. This is not the feature that should bring us there. As the current implementation does, manipulating `--embed-dir=` similar to the way `#include` is handled alongside `-I` flags is a much better use of not only implementer time, but programmer time. It fits perfectly into the mental model ("include, but for resources") and lets users keep their file names as the unit of interchange as far as names go.

C++ does not need to make for itself a reputation of trying to be an extremely unique snowflake at the cost of usability and user friendliness.

§ 9. Acknowledgements

Thank you to Hana Dusíková for helping jumpstart and kick off the final public Clang Implementation / Patches and the newer design ideas for C++29.

A big thank you to Andrew Tomazos for replying to the author's e-mails about the prior art. Thank you to Arthur O'Dwyer for providing the author with incredible insight into the Committee's previous process for how they interpreted the Prior Art.

A special thank you to Agustín Bergé for encouraging the author to talk to the creator of the Prior Art and getting started on this. Thank you to Tom Honermann for direction and insight on how to write a paper and apply for a proposal.

Thank you to Arvid Gerstmann for helping the author understand and use the link-time tools.

Thank you to Tony Van Eerd for valuable advice in improving the main text of this paper.

Thank you to Lilly (Cpplang Slack, @lillypad) for the valuable bikeshed and hole-poking in original designs, alongside Ben Craig who very thoroughly explained his woes when trying to embed large firmware images into a C++ program for deployment into production. Thank you to Elias Kounen and Gabriel Ravier for wording review.

For all this hard work, it is the author's hope to carry this into C++. It would be the author's distinct honor to make development cycles easier and better with the programming language we work in and love. ♥

§ References

§ Informative References

[CIRCLE-EMBED-TWEET]

Sean Baxter. *@embed added to Circle*. URL: <https://twitter.com/seanbax/status/1205195567003045888>

[CLANG-LARGE-INIT-BUG]

LLVM Foundation. *Memory Consumption Reduction for Large Array Initialization?*. URL: https://bugs.llvm.org/show_bug.cgi?id=44399

[CONSTEXPR-ALL-THE-THINGS]

Ben Deane; Jason Turner. *constexpr All The Things: CppCon 2017*. September 25th, 2017. URL: <https://www.youtube.com/watch?v=PJwd4JLYJJY>

[eElipsis]

Jens Gustedt. *eElipsis, a language independent preprocessor*. April 28th, 2026. URL: <https://gustedt.gitlabpages.inria.fr/ellipsis>

[GCC-LARGE-INIT-BUG-C]

GCC. *[8/9/10 regression] Uses lots of memory when compiling large initialized arrays*. URL: https://gcc.gnu.org/bugzilla/show_bug.cgi?id=12245

[GCC-LARGE-INIT-BUG-CPP]

GCC. *[8/9/10 regression] Uses lots of memory when compiling large initialized arrays*. URL: https://gcc.gnu.org/bugzilla/show_bug.cgi?id=14179

[INCBIN]

Dale Weiler (graphitemaster). *incbin: load files at 'assembly' time*. URL: <https://github.com/graphitemaster/incbin>

[N4778]

Richard Smith. *Working Draft, Standard for Programming Language C++*. 8 October 2018. URL: <https://wg21.link/n4778>

[NONIUS-VISUAL-C-ERROR]

R. Martinho Fernandes. *nonius generated HTML Reporter*. September 1st, 2016. URL: https://github.com/libnonius/nonius/blob/devel/include/nonius/reporters/html_reporter.h%2B%2B#L42

[P0373R0]

Andrew Tomazos. *Proposal of File Literals*. 21 May 2016. URL: <https://wg21.link/p0373r0>

[P0466R1]

Lisa Lippincott. *Layout-compatibility and Pointer-interconvertibility Traits*. 12 February 2018. URL: <https://wg21.link/p0466r1>

[P1130]

JeanHeyd Meneide. *Module Resource Requirement Propagation*. November 26th, 2018. URL: https://theohd.dev/_vendor/future_cxx/papers/d1130.html

[P1130R1]

JeanHeyd Meneide. *Module Resource Requirement Propagation*. 21 January 2019. URL: <https://wg21.link/p1130r1>

[P1130R2]

JeanHeyd Meneide. *Module Resource Requirement Propagation*. 27 March 2026. URL: <https://wg21.link/p1130r2>

[P1967]

JeanHeyd Meneide. *Module Resource Requirement Propagation*. February 14th, 2025. URL: https://theohd.dev/_vendor/future_cxx/papers/d1967.html